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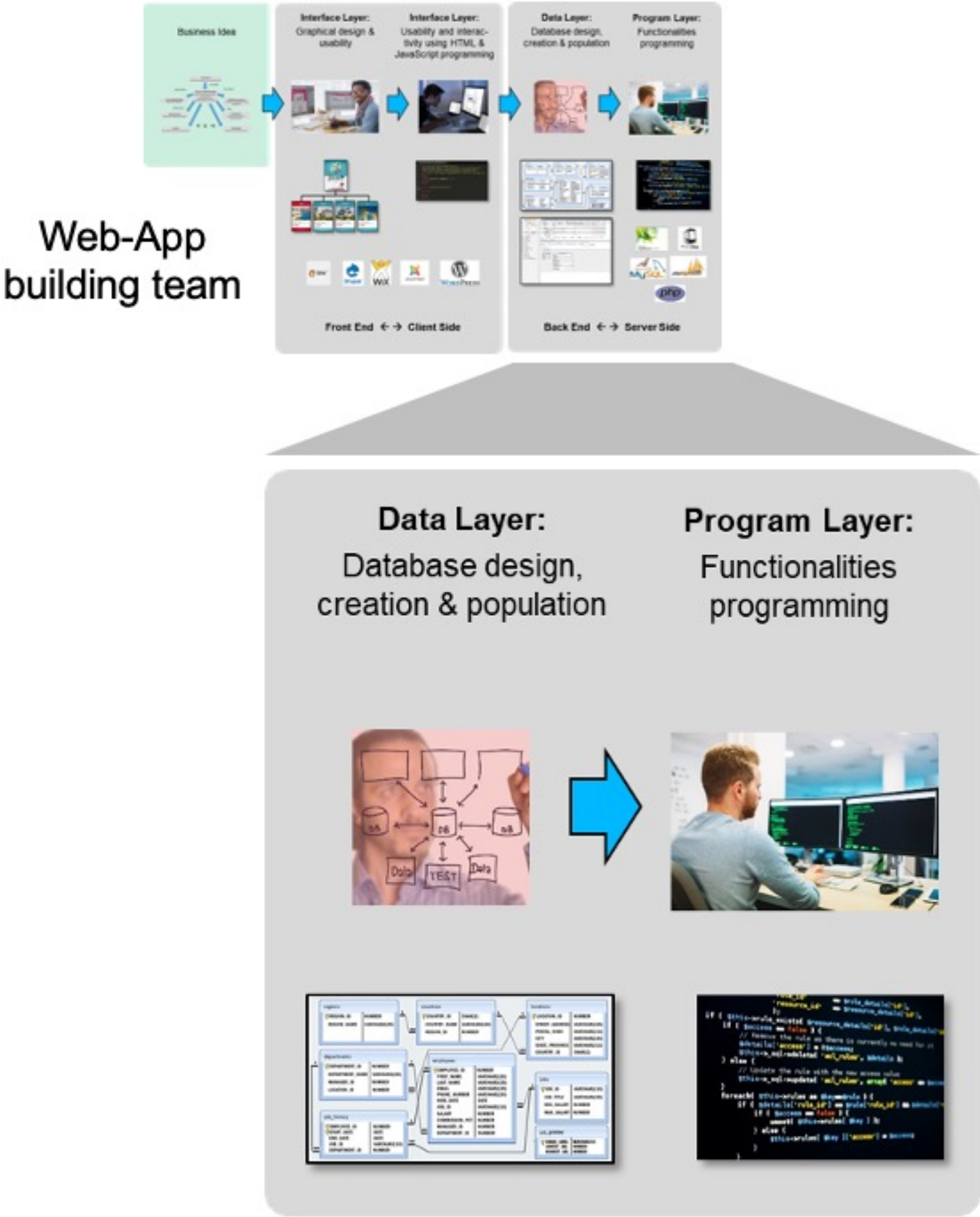
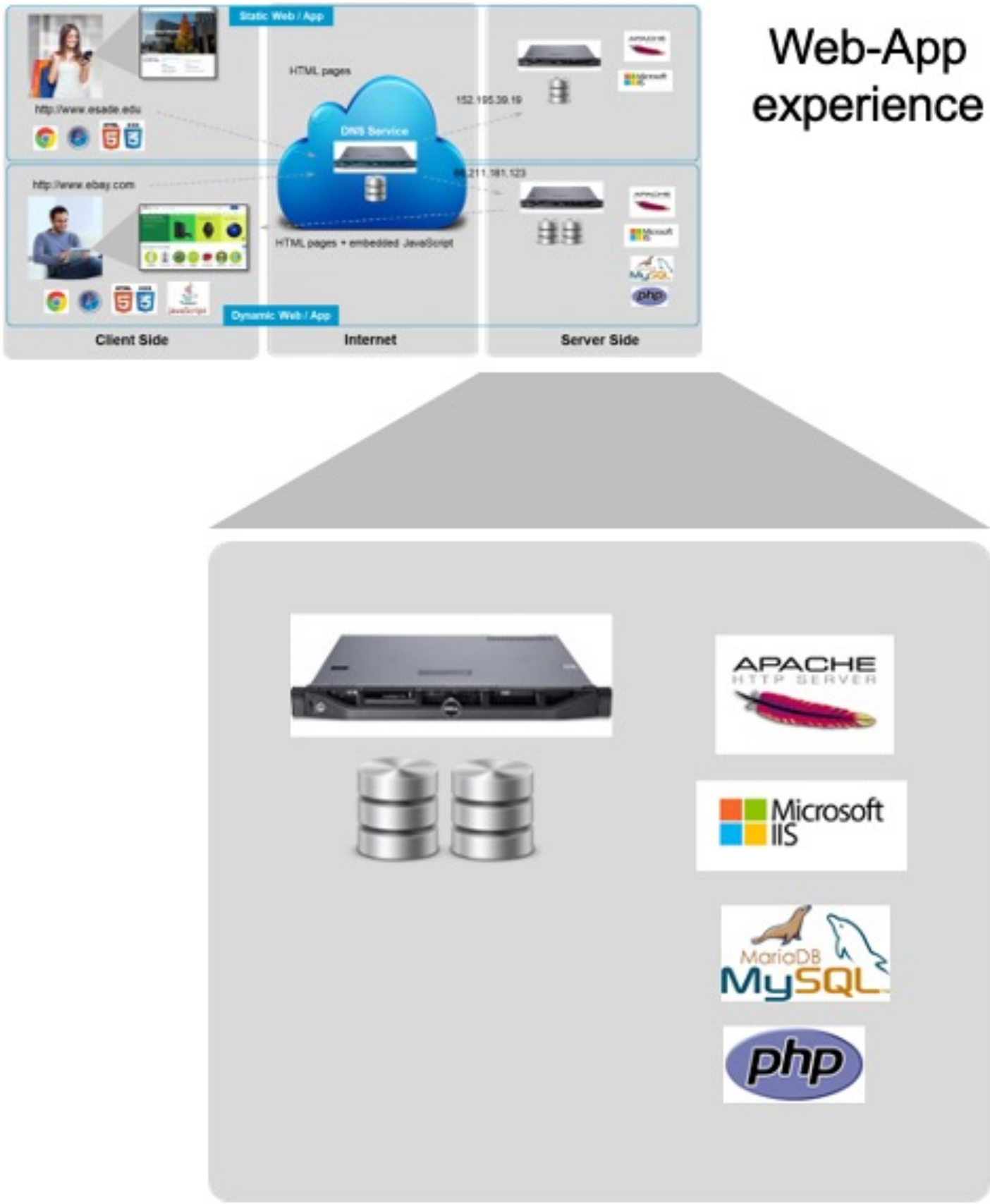
# Development Tools for Web Apps

## Back End Web-App development - 1

### Sessions 6, 7, 8

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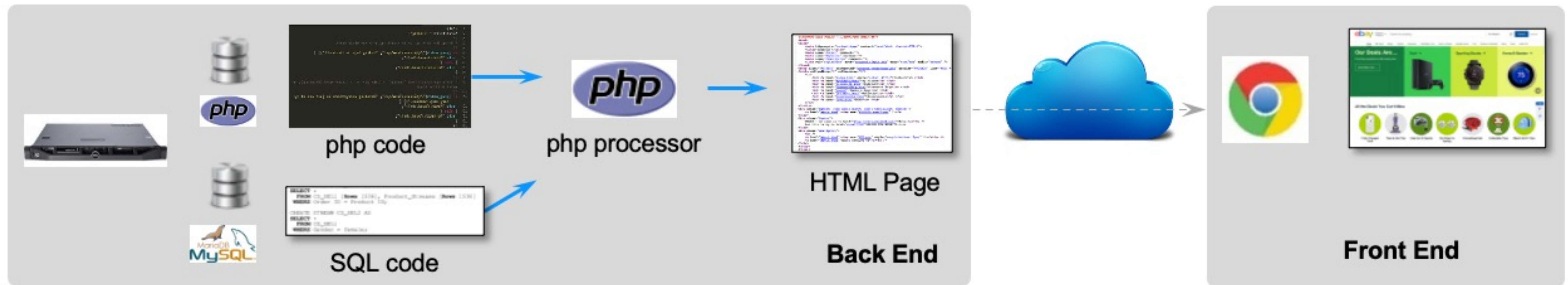
Back End: data & program layers



# PHP

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- **PHP** is a general purpose programming language designed for web development in the back end (server side), typically having SQL instructions embedded for database data manipulation.



- PHP instructions are embedded in HTML pages.
- PHP processor takes the HTML page to be presented and executes (interprets) the PHP code, to add the dynamic information in it. PHP can retrieve information from a database (having SQL code embedded) and process the information to be included in the HTML page before send to the browser (user).
- PHP processor transforms the PHP code and only sends to the browser HTML instructions (possibly having included JavaScript instructions).

*Creating Dynamic Web Pages*

**3rd Edition**

*Programming*

**PHP**

*Kevin Tatroe, Peter MacIntyre  
& Rasmus Lerdorf  
Foreword by Michael Bourque*

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## English



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## PHP statements

- `$mail = john.doe@esade.edu // defines $mail variable`
- `echo("xxxxx"); // it shows the text xxxxx in the built HTML page`
  - Ex. `echo("<h1>Hello</h1>");`
- `exit(); // it stops executing the program (building the HTML page)`
- `die("xxxxx"); // it is equal to : echo("xxxxx"); exit();`
- `session_start(); // start the session to use session variables`
- `$_SESSION['userid'] = $user; // Defines the session variable 'userid'`
- `$id_user = $_SESSION['userid']; // Retrieve userid from the Session variable defined in the previous php page`

## Managing MySQL with PHP

```
// Define connection variables
$host = "localhost";
$db_username = "root";
$db_password = "";
$db_name = "mydatabase";

// Connect to the Database. Store the connection in $conn

$conn = mysqli_connect ($host, $db_username, $db_password, $db_name) or die ('Connection Error');

// execute query and store the list of users into a recordset $rs
$rs = mysqli_query($conn, 'SELECT * FROM users');
```

```
mysqli_num_rows($rs) // Number of rows retrieved by query
```

```
$user = mysqli_fetch_row($rs); // fetch 1st row of the recordset
```

| userID | name          | password | Gender | email             | conditions | prefixphone | phone    |
|--------|---------------|----------|--------|-------------------|------------|-------------|----------|
| 6      | Peter Gabriel | peter    | M      | peter@gabriel.com | Y          | 44          | 95555888 |

```
$user[0]; $user[1]; $user[2]; $user[3]; .....
```

```
mysqli_data_seek($rs, 0); // positions the pointer on the first record
```

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## HTML Forms and PHP

new\_user.html

```
<form method="POST" action="user_reg.php">
<h4>Form Register</h4>
  Name : <input type="text" name="firstname" >
  Email: <input type="text" name="email" >
  <input type="submit" value="Sing Up">
</form>
```

User\_reg.php

```
<?php
$f_name = $_POST['firstname'];
$e_mail = $_POST['email'];

echo "Hello $fname";

?>
```

|                                        |                                             |
|----------------------------------------|---------------------------------------------|
| Name:                                  | <input type="text" value="John"/>           |
| Email:                                 | <input type="text" value="john@esade.edu"/> |
| <input type="button" value="Sign Up"/> |                                             |

```
$f_name ->John
$e_mail ->John@esade.edu
```

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## Connection to the database

Each php page must establish a connection to the database.

db.php

```
<?php
    session_start(); // start a new session

    $conn = mysqli_connect(        // connect to the database

        'localhost',              // host (machine) where the database is located. In our case it is our local server
        'root',                   // user with permission to access the database
        'root',                   // password of this user
        'mkt_tasks'               // database name

    ) or die(mysqli_errno($mysqli)); // End of program if conex not successfull

?>
```

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# Development Tools for Web Apps

## Login & Register

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## Connecting to the database

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`<?php` *Starts a session*  
`session_start();` *It's like opening a temporary folder to store user info while they browse.*

```
$conn = mysqli_connect(
    'localhost', // database server
    'root',      // username
    'root',      // password
    'tasks'      // database name
) or die(mysqli_error($mysqli));
```

### Connects to the database

“Hello database, I want to talk to you.”

**'localhost'**: the database is on the same server.  
**'root'**: username (like your login name).  
**'root'**: password (like your key).  
**'tasks'**: the database we want to use.

### What if it fails?

**or die(...)** shows an error if something goes wrong



*db.php*

### What Is It For?

It allows the app to **save, read, or update tasks**.  
 It's the **bridge** between your website and your data.



*Database server*

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Login

Return

if the user exists, the list of his tasks is displayed.

| Tasks List                            |       |                          |       |                     |                                                                                                                                                                         |
|---------------------------------------|-------|--------------------------|-------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TaskID                                | Title | Description              | User  | Created at          | Action                                                                                                                                                                  |
| 3                                     | id01  | Repair Millennium Falcon | han   | 2021-06-10 16:16:54 |   |
| 4                                     | Rp36  | Flight Tatoonie          | han   | 2021-06-10 16:17:15 |   |
| 5                                     | DE45  | Laser combat training    | luke  | 2021-06-10 16:17:52 |   |
| 6                                     | FT12  | Meditation day           | luke  | 2021-06-10 16:18:24 |   |
| 7                                     | AD01  | Systema maintenance      | admin | 2021-06-10 16:18:44 |   |
| 8                                     | AD02  | Export database          | admin | 2021-06-10 16:18:57 |   |
| <div>Return</div> <div>New Task</div> |       |                          |       |                     |                                                                                                                                                                         |

Tasks\_list.php

Error. Uses doesn't exist in the database and/or you must fill all fields

Return

Error\_login.html

Login

Write your username

Write your password

Login

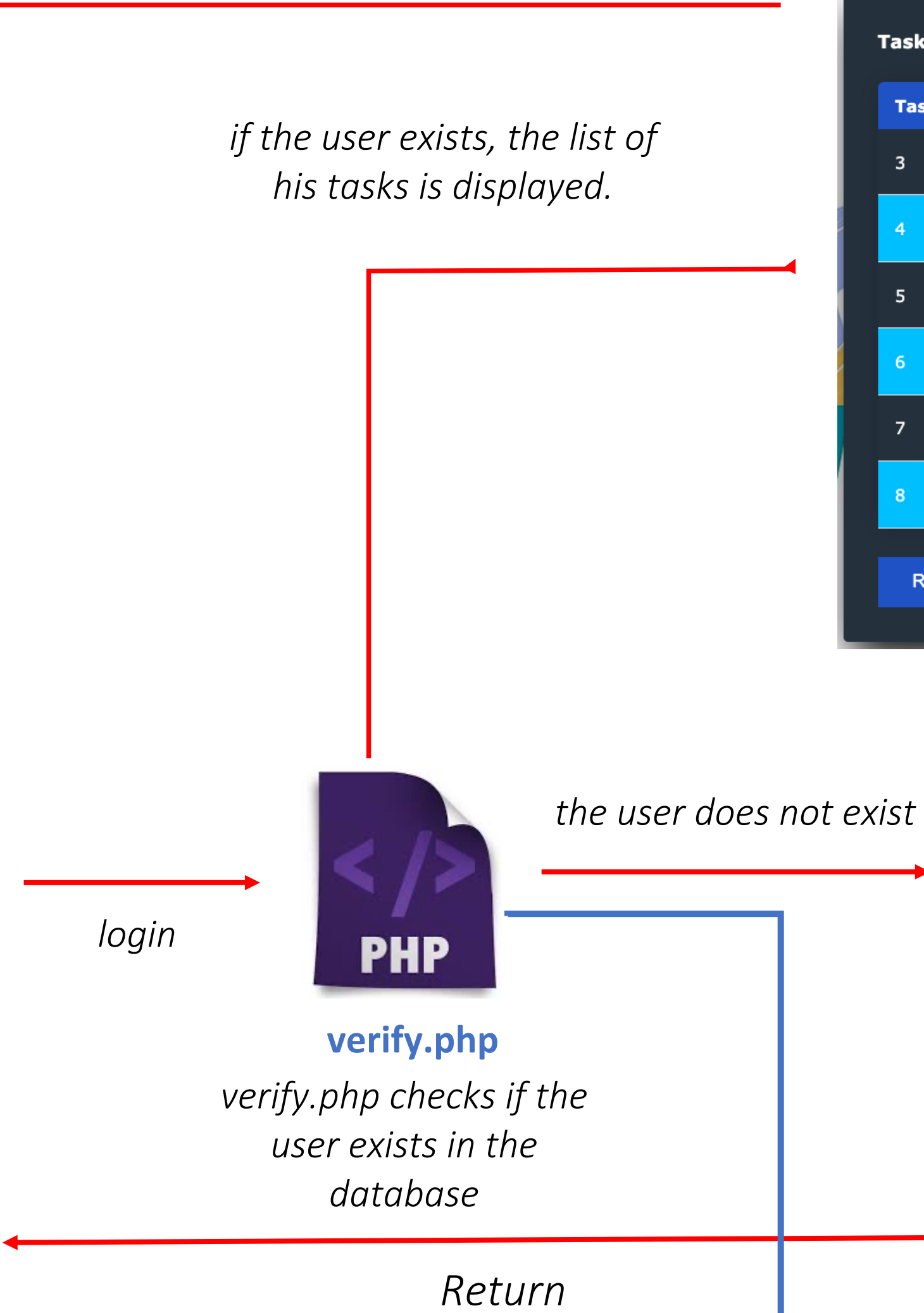
Cancel

login.html



verify.php

verify.php checks if the user exists in the database



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Register new user

Return

Error. User already exist in the database

Return

User\_exist.html

## Form Register

Write your firstname

Write your lastname

Write your username

Write your password

Register

¿Already have an account?

new\_user.html

Register



User\_reg.php

the user already exist

the new user is inserted in the database  
and , he/she can add new tasks

Return

Cancel

Already have an account

## Tasks List

| TaskID | Title | Description | User | Created at | Action |
|--------|-------|-------------|------|------------|--------|
|--------|-------|-------------|------|------------|--------|

Return

New Task

tasks\_list.php

## Login

Write your username

Write your password

Login

Cancel

login.html



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